Task Title: "Movie Explorer – Discover Your Favorite Films"

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Project Story:

You have been hired by a startup that to create a **Movie Explorer App**. This web application will allow users to search for movies, view details, and discover trending films. The app will fetch real-time data from the **TMDb (The Movie Database) API** to display information about movies.

Features & Requirements:

1. **User Interface (UI):**

- User Login interface with username and password.
- A **search bar** where users can type a movie name to get relevant results.
- Display a **grid of movie posters**, each showing the title, release year, and rating.
- Clicking on a movie should open a **detailed view** with additional information (overview, genre, cast, trailer link, etc.).
- A **trending movies section** displaying popular movies from the API.
- Implement a **light/dark mode** for better user experience.

2. **API Integration:**

- Use **The Movie Database (TMDb) API** (https://developers.themoviedb.org/3) to fetch:
  - Trending movies
  - Movie search results
  - Movie details (title, poster, description, rating, genres, etc.)
- Implement **infinite scrolling** for search results.
- Handle API errors gracefully with user-friendly messages.

3. **State Management:**

- Use **React Context API or Redux** to manage movie data.

- Store the **user's last searched movie** in local storage for persistence.
- Allow users to **save favorite movies** to a list stored locally.

4. **Extra Features (Bonus for Enthusiastic Interns! ☺):**

- Allow users to filter movies by **genre, year, or rating**.
- Show **YouTube trailers** using the YouTube API or an embed link from TMDb.
- Implement a **"Load More" button** instead of infinite scroll for better UX.

Technical Guidelines & Instructions:

1. **Setup & Environment:**
    - Create a new React app using Create React App.
    - Use axios for API requests.
    - Install and use **Material-UI (MUI)** for styling.
2. **Development Process:**
    - Break down the UI into **reusable components** (e.g., MovieCard, SearchBar, MovieDetails).
    - Implement **React Router** for navigation (Home, Movie Details, Favorites).
    - Follow **mobile-first responsive design** principles.
3. **Testing & Deployment:**
    - Deploy the app using **Vercel or Netlify** and provide a live demo link.
4. **Deliverables:**
    - **GitLab Repository** with well-structured, commented code.
    - **README.md** explaining project setup, API usage, and features implemented.
    - **Deployed live link** for the working app.